Sales Promotion and Channel Management under Uncertainty

Monte Carlo Simulation, Inventory Decisions, and Channel Incentives

Business Modeling & Simulation / Operations Analytics

Portfolio Project Report

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Executive Summary

This project examines two related **inventory decisions** in which firms must commit supply before uncertain market outcomes are realized. The first case studies **a catalog furniture promotion with uncertain demand and selling price**. The second studies **Flextrola**, an electronics systems integrator that must **place a single pre-season component order** with supplier Solectrics. Both cases are modeled in Excel using **Monte Carlo simulation**, **probability distributions**, Named Ranges, **conditional logic**, **confidence intervals**, **What-If Data Tables**, and **decision-search logic**.

In the saved workbook snapshot, the Sales Promotion base policy of S = 3,000 produces an estimated mean profit of $97,546, a 95% confidence interval of $91,557 to $103,534, and a 2.1% stockout probability. Across the tested inventory grid, the highest saved simulated expected profit occurs at S = 1,900, approximately $125,644. In the Flextrola base case, S = 1,200 yields an estimated stockout probability of 36.3%, mean fill rate of 91.6%, and expected Flextrola profit of $34,335, with a 95% confidence interval of $33,578 to $35,092. Across the Flextrola expected-profit Data Table, the highest saved simulated mean profit occurs at S = 1,400, at approximately $34,516.71. Separately, Solectrics' supplier-profit block reaches its maximum at S = 1,600, corresponding to $32,000 in supplier profit. Taken together, the cases show that inventory decisions under uncertainty require balancing expected profit, service performance, overstock exposure, and—in a multi-firm channel—the allocation of economic risk.

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1. Business Problem and Analytical Motivation

Many commercial decisions must be made before managers know the demand, price, or market conditions that will ultimately determine financial performance. **Inventory decisions** are especially exposed to this timing problem: ordering too little can create stockouts and lost full-price sales, whereas ordering too much creates excess inventory that must be discounted or liquidated. The decision is therefore not to eliminate uncertainty, but to choose a policy that performs well across plausible market realizations.

The project uses two cases to develop this idea progressively. **The Sales Promotion case** is a single-firm inventory decision with uncertainty in both demand and selling price. **The Flextrola case** retains the single-period ordering structure but adds a supplier–integrator relationship, service metrics, salvage value, and potentially different incentives across channel members. This structure mirrors the broader operations-research view of simulation optimization: a decision is evaluated through repeated stochastic outcomes, and alternatives are compared using estimated objective values rather than a single deterministic forecast.

Spreadsheet simulation is particularly useful for this class of problems because the operational logic, random inputs, outputs, and decision comparisons can be kept visible and auditable. Published operations-research work likewise uses spreadsheet simulation and iterative search to support inventory decisions and emphasizes simulation as a practical way to compare policies when performance is stochastic.

2. Theoretical and Methodological Framework

2.1 Decision Making under Uncertainty

Let S denote a feasible pre-season order quantity and let Y represent uncertain market conditions. For any realized state Y, the model produces an economic outcome g(S,Y). The decision objective is to compare the expected performance V(S)=E[g(S,Y)] across feasible values of S. Because the expectation is not evaluated analytically in the workbook, it is estimated by Monte Carlo simulation.

For N simulated seasons, the expected outcome is approximated by the sample mean: $V(S) \approx (1/N) \Sigma\, g(S,Y_i)$. The same simulated seasons can also be used to estimate probabilities such as stockout risk, service measures such as fill rate, and uncertainty around the estimated mean through a confidence interval.

2.2 Monte Carlo Simulation

Each simulation run generates random inputs from the assumed probability distributions, maps those inputs through the business logic, and records the resulting outputs. Repeated runs create an empirical distribution of profit and service outcomes. The workbook uses RAND() as the underlying random-number generator and inverse distribution functions such as NORM.INV() to transform uniform random draws into demand realizations. This is consistent with standard spreadsheet Monte Carlo practice, where random functions and inverse probability distributions are combined to generate stochastic business inputs.

2.3 Confidence Intervals and Simulation Precision

A Monte Carlo mean is an estimate rather than an exact deterministic value. The workbook reports a 95% confidence interval for mean profit using the familiar normal approximation, mean $\pm$ 1.96 $\times$ s/$\sqrt{N}$, where s is the sample standard deviation. Increasing N reduces the standard error approximately at the 1/$\sqrt{N}$ rate. The Flextrola case therefore uses 5,000 simulated seasons, compared with 1,000 in the Sales Promotion case, to obtain greater numerical precision for a richer set of performance measures.

2.4 Sensitivity Analysis and Simulation-Based Decision Search

After evaluating a base order quantity, the workbook uses one-way Excel Data Tables to repeat the model over a grid of candidate order quantities. This converts the simulation from a descriptive exercise into a decision-support model: each S is associated with simulated profit and service outcomes, and the saved workbook uses MAX/LARGE, MATCH, and INDEX logic to identify the best value on the tested grid. Because RAND()-based models recalculate, small differences between alternatives can reflect simulation noise; the decision region and economic trade-offs are therefore more informative than treating a small point-estimate difference as exact.

3. Case 1 — Sales Promotion Inventory Decision

3.1 Business Context and Assumptions

A catalog company plans a two-month furniture promotion one year in advance and must commit its chair inventory before the selling season begins. Demand and selling price are uncertain because of market conditions, fashion trends, advertising effectiveness, and competitors' actions. The workbook models demand as Normal with mean 2,000 and standard deviation 500, truncated at zero, and selling price as Uniform from $200 to $300. Each chair costs $175. Unsold units can be sold after the promotion at 50% of the realized selling price. The base policy is S = 3,000 and the simulation uses N = 1,000 seasons.

3.2 Excel Model Logic and Project Steps

1. Define the business inputs and decision variable using Named Ranges: demand parameters, price bounds, unit cost, order quantity S, and simulation size.

2. Generate seasonal demand with INT(MAX(NORM.INV(RAND(),mu,sigma),0)) and price with a+(b-a)×RAND().

3. For each simulated season, calculate sales as the smaller of demand and supply, then calculate sales revenue, leftover inventory, discount revenue, procurement cost, total profit, and a stockout indicator.

4. Aggregate the simulated seasons into expected profit, a 95% confidence interval for mean profit, and stockout probability.

5. Construct the profit histogram to examine the distribution rather than relying only on the mean.

6. Use a one-way Data Table to evaluate S = 500, 600, …, 3,500 and compare expected profit with stockout probability.

7. Use the saved decision-search block to identify the highest simulated expected profit on the tested grid.

3.3 Results

Metric	Saved Workbook Result	Interpretation
Base order quantity	3,000 chairs	Reference pre-season inventory policy
Expected profit	$97,545.77	Average profit across 1,000 simulated seasons
95% CI for mean profit	$91,557.39 – $103,534.14	Sampling uncertainty around the estimated mean
Stockout probability	2.1%	Estimated probability demand exceeds

		supply
Saved grid maximum	S = 1,900; $125,644.28	Highest saved simulated expected profit on the tested grid

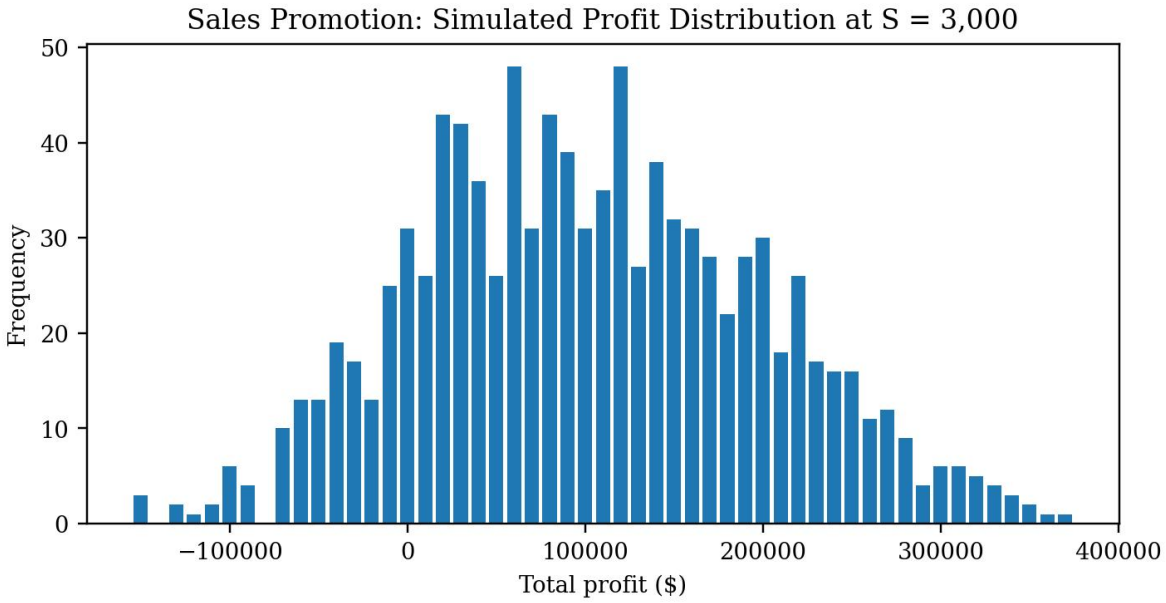


Figure 1. Sales Promotion simulated profit distribution at the base order quantity.

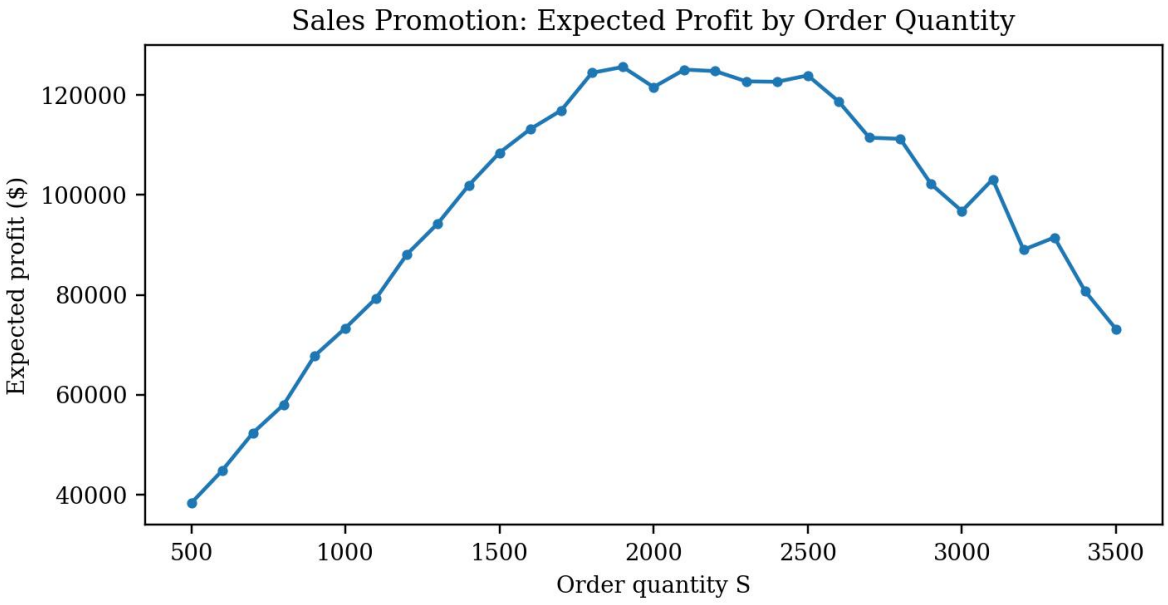


Figure 2. Expected profit by order quantity using the workbook Data Table.

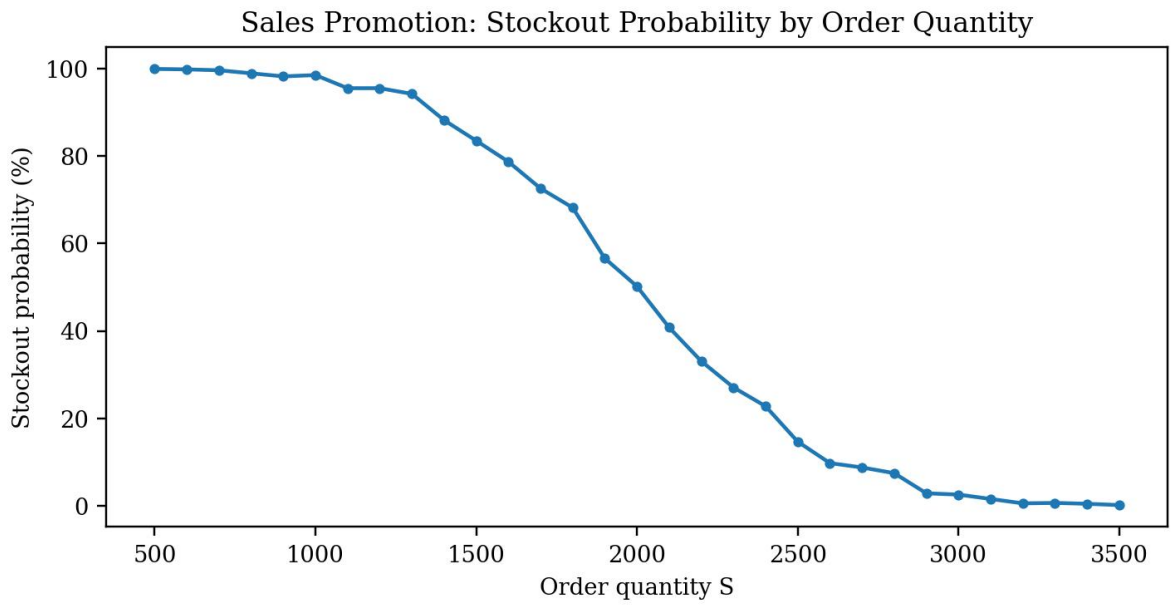


Figure 3. Stockout probability by order quantity using the workbook Data Table.

3.4 Interpretation and Business Implications

The base policy of S = 3,000 is service-oriented: its saved stockout probability is only 2.1%. However, the saved expected profit of $97,546 is below the highest simulated value on the tested grid, which occurs at S = 1,900 with approximately $125,644. The sensitivity pattern therefore demonstrates that minimizing stockouts is not the same objective as maximizing expected profit.

Economically, **additional inventory** has two opposing effects. It protects full-price sales when demand is high, but it also increases the number of units exposed to the discounted post-promotion salvage channel when demand is lower than expected. The expected-profit curve rises from very low order quantities, reaches a broad high-profit region, and then generally declines as excess-inventory exposure becomes more important. Meanwhile, stockout probability falls sharply with inventory. The decision is therefore a profit–service trade-off rather than a simple rule to maximize availability.

For management, the result argues for evaluating promotional inventory with both economic and service metrics. A manager focused only on avoiding stockouts could over-order and sacrifice expected profit, while a manager focused only on the single highest simulated mean could overstate the precision of the simulation. The useful managerial conclusion is the location of the attractive decision region and the economic mechanism that creates it.

4. Case 2 — Flextrola Channel Inventory Decision

4.1 Business Context and Channel Economics

Flextrola is an electronic systems integrator preparing a key component for a next-generation product with supplier Solectrics. Short product life cycles and long supplier lead times create a single-order problem: Flextrola has one opportunity to order before the selling season, cannot revise the order later, and cannot return excess components to Solectrics. Unsold units must be liquidated in a secondary electronics market.

Seasonal demand is modeled as Normal with mean 1,000 and standard deviation 600, with nonnegative truncation in the implemented demand formula. Solectrics incurs a production cost of $52 and charges

Flextrola $72 per component. Flextrola sells the completed unit for $121 and can recover $50 for leftover inventory in the secondary market. The base order is S = 1,200, the simulation uses 5,000 seasons, and the tested contractual menu runs from 1,000 to 1,600 units in increments of 100.

4.2 Excel Model Logic and Project Steps

1. Generate nonnegative seasonal demand using MAX(0,NORM.INV(RAND(),mean,sigma)).

2. For each season, calculate regular-market sales, sales revenue, leftover inventory, secondary-market revenue, procurement cost, and Flextrola profit.

3. Calculate service measures including a stockout indicator and fill rate, together with gross-margin percentage.

4. Aggregate 5,000 seasons into expected sales, expected leftover inventory, mean fill rate, expected gross margin, expected Flextrola profit, and a 95% confidence interval.

5. Use a Data Table to compare Flextrola profit and confidence intervals across the order menu.

6. Separately evaluate Solectrics' profit across the same order quantities to expose channel-level incentive differences.

7. Use the workbook's saved maximum blocks to summarize the preferred order quantity under the modeled objective.

4.3 Results

Metric	Saved Workbook Result	Interpretation
Base order quantity	1,200 units	Reference single pre-season order
Stockout probability	36.3%	Probability demand exceeds available units
Expected sales	855.4 units	Average units sold in the regular market
Expected leftover	344.6 units	Average units sent to the secondary market
Mean fill rate	91.6%	Average fraction of demand satisfied
Expected gross margin	23.3%	Average simulated gross-margin percentage
Expected Flextrola profit	$34,334.86	Average seasonal profit at S = 1,200
95% CI for mean profit	$33,577.73 – $35,091.98	Sampling uncertainty around mean profit
Solectrics profit at base S	$24,000.00	Supplier objective at S = 1,200
Flextrola saved profit maximum	S = 1,400; $34,516.71	Highest saved simulated mean_profit on the tested order grid
Solectrics saved profit maximum	S = 1,600; $32,000.00	Highest saved S_profit for the supplier on the tested order grid

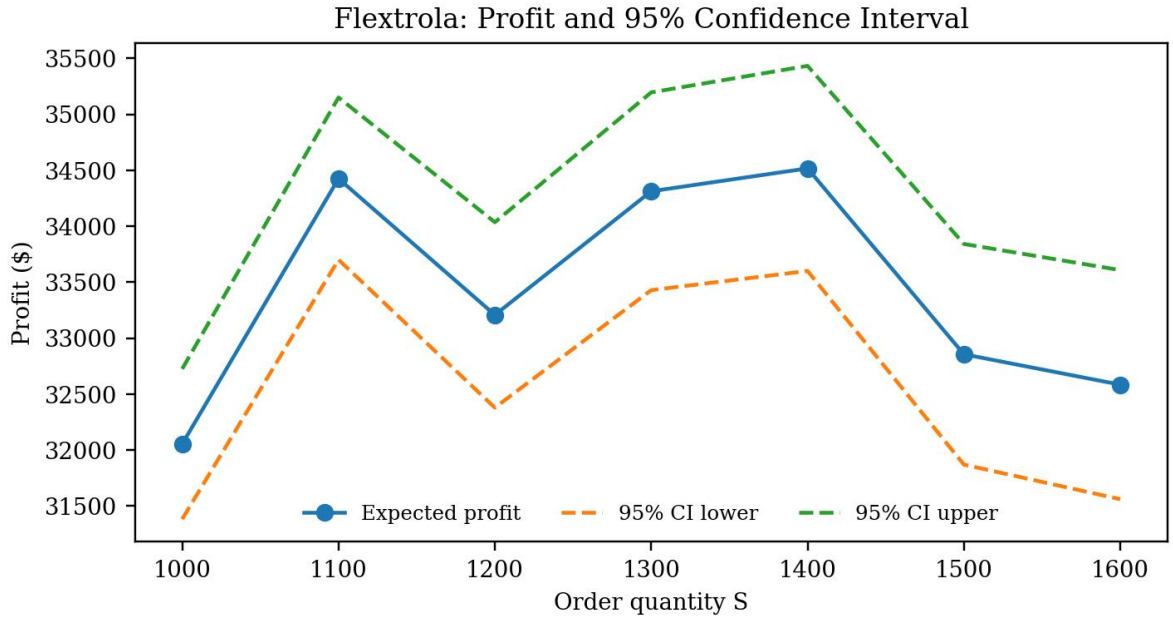


Figure 4. Flextrola expected profit and 95% confidence interval across the tested order menu.

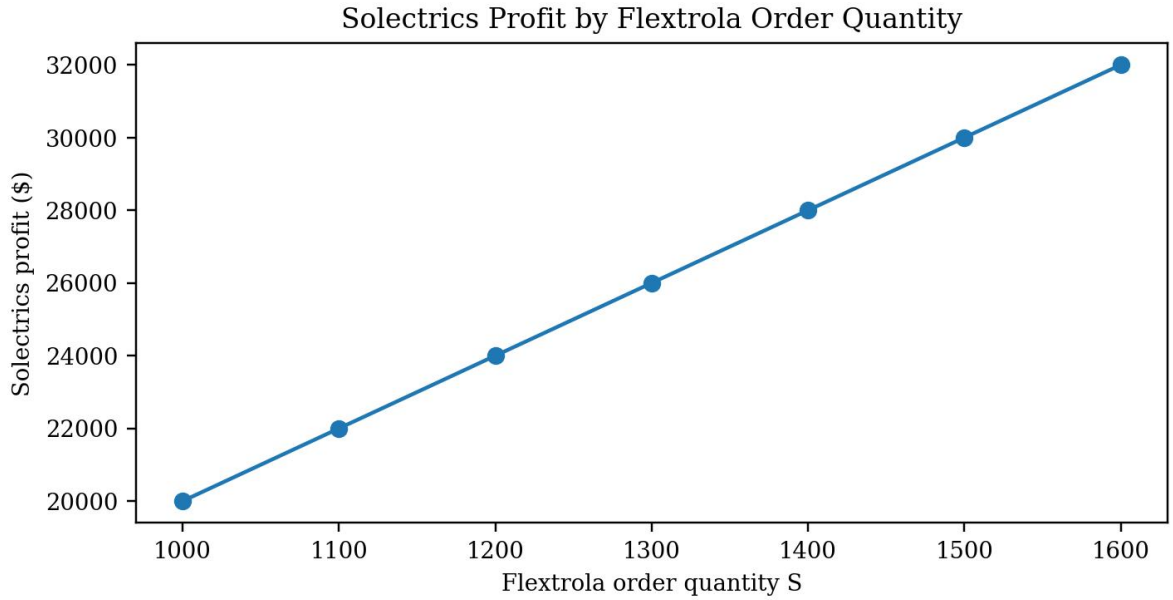


Figure 5. Solectrics profit by Flextrola order quantity.

4.4 Interpretation and Channel Implications

At the base order S = 1,200, the model estimates a 36.3% stockout probability while maintaining a 91.6% mean fill rate. Expected regular-market sales are 855.4 units and expected leftover inventory is 344.6 units. Expected Flextrola profit is $34,335, with a 95% confidence interval of approximately $33,578 to $35,092. These metrics show why a single service measure can be misleading: the model can have a substantial probability of some stockout while still satisfying a high fraction of total demand on average.

Across the Flextrola expected-profit Data Table, the highest saved simulated mean profit occurs at S = 1,400, at approximately $34,516.71. This is Flextrola's simulated profit optimum on the tested order grid. In contrast, Solectrics' supplier profit is maximized at S = 1,600, where supplier profit equals $32,000. The difference is

economically meaningful: the two channel members evaluate the same order quantity using different profit functions. Each Data Table evaluation is generated from RAND()-based stochastic recalculation, so nearby alternatives can move relative to one another when the workbook is recalculated. The most defensible interpretation is therefore to identify the economic trade-off and decision region rather than treat a small simulated difference as an immutable optimum.

The channel structure adds a second layer of managerial interpretation. Solectrics earns $20 per ordered component ($72 transfer price minus $52 production cost), so its modeled profit rises mechanically with Flextrola's committed order. Flextrola, by contrast, bears the downside of excess inventory because a unit purchased for $72 recovers only $50 if it remains unsold. The supplier therefore benefits from larger committed volume, while the integrator must balance service against overstock exposure. This incentive asymmetry is a channel-management issue, not merely an inventory calculation.

In practice, such misalignment motivates attention to contract design and risk sharing. Buyback provisions, quantity flexibility, revenue sharing, or other contractual mechanisms can alter who bears the cost of unsold inventory. The present workbook does not optimize those contracts; rather, it demonstrates why a decentralized supply relationship can produce different preferred decisions for different parties.

5. Cross-Case Findings

The two cases share the same analytical backbone but emphasize different managerial dimensions. Both begin with a pre-season quantity decision, generate uncertain demand, translate realized demand into sales and leftovers, and evaluate performance over many simulated seasons. The Sales Promotion case adds uncertain selling price and emphasizes the trade-off between expected profit and stockout probability. The Flextrola case adds a supplier relationship, secondary-market salvage, fill rate, gross margin, and separate channel-member economics.

Dimension	**Sales Promotion**	**Flextrola**
Decision	Promotional chair inventory	Single pre-season component order
Main uncertainty	Demand and selling price	Demand
Simulation size	1,000 seasons	5,000 seasons
Service measure	Stockout probability	Stockout probability and fill rate
Overstock recovery	50% of realized selling price	$50 secondary-market price
Decision analysis	S = 500 to 3,500	S = 1,000 to 1,600
Additional managerial issue	Profit–service trade-off	Channel incentive alignment

6. Simulation Convergence, Reliability, and Limitations

6.1 Sampling Variability

The workbook is intentionally stochastic. RAND()-based formulas mean that pressing F9, reopening the workbook in a recalculating application, or recalculating a What-If Data Table can generate a new Monte Carlo sample. Consequently, the base-case output and a Data Table row using the same nominal order quantity need not be numerically identical if they were produced in different recalculations. This is simulation variation, not necessarily a formula error.

Confidence intervals help quantify the precision of estimated mean profit, but they should not be confused with the full distribution of business outcomes. They measure sampling uncertainty around the estimated mean. They also do not prove that the order quantity with the highest simulated point estimate is the unique true optimum, particularly when neighboring alternatives have similar estimated values.

6.2 Model Limitations

- The probability distributions and their parameters are treated as known rather than estimated from a documented historical dataset.
- Demand and market outcomes are represented by stylized distributions and independent simulated seasons.
- The models abstract from operational constraints beyond the stated inventory decision, such as capacity, financing, lead-time variability, and multi-period replenishment.
- The Sales Promotion model assumes leftover chairs can all be liquidated at exactly half of the realized selling price.
- The Flextrola model assumes a fixed transfer price, fixed secondary-market value, no returns, and a single pre-season order.
- Data Table rankings can change across recalculations because the model uses random sampling; more replications or variance-reduction techniques would be appropriate if very small differences between candidate policies were decision-critical.

7. Managerial and Analytical Implications

The central managerial lesson is that **uncertainty** should be incorporated into the decision process rather than collapsed into a single forecast. A deterministic 'expected demand' calculation cannot show the probability of stockout, the distribution of profit, the economic cost of leftover inventory, or the sampling uncertainty around estimated performance. Monte Carlo simulation makes those dimensions visible.

The project also shows why decision quality depends on the **objective** being optimized. A service-oriented policy may carry more inventory than the profit-maximizing region; a supplier may prefer a larger committed order than the downstream firm; and a point estimate that appears best in one simulation run may not remain best after recalculation. For business analytics, this shifts the focus from producing one number to understanding the structure of the decision, the sensitivity of outcomes, and the trade-offs that management must explicitly choose.

From an implementation perspective, the workbook demonstrates a transparent spreadsheet architecture: inputs and assumptions feed a stochastic simulation block; simulation outputs are summarized statistically; Data Tables automate repeated policy evaluation; and charts translate the numerical results into decision-relevant patterns. This structure makes the model auditable and suitable for communicating analytical reasoning to nontechnical stakeholders.

8. Conclusion

This project develops two spreadsheet-based Monte Carlo models for **inventory decisions under uncertainty**. **The Sales Promotion case** demonstrates how uncertain demand and price jointly shape promotional profit and stockout risk, while **the Flextrola case** extends the framework to a supply-channel setting with service performance, salvage value, and distinct incentives for the integrator and supplier. Across both cases,

simulation converts uncertain inputs into distributions of operational and financial outcomes, and Data Tables connect those outcomes to actionable order-quantity decisions.

The analysis supports a broader conclusion: the most useful role of simulation is not to claim certainty about a single optimal number, but to quantify uncertainty, expose the economics of alternative decisions, and identify where managerial trade-offs occur. **That combination of stochastic modeling, statistical interpretation, sensitivity analysis,** and **business reasoning** is the core analytical contribution of the project.

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